



SURGICAL SKILL ASSESSMENT OF CATARACT VIDEOS



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INTRODUCTION

Enhancing Surgical Skill Assessment

Surgical Skill
Affects
Outcomes



Surgical Skill
Assessment &
Feedback are
Subjective &
Inconsistent



AI Enables
Objective &
Consistent
Assessment
& Feedback



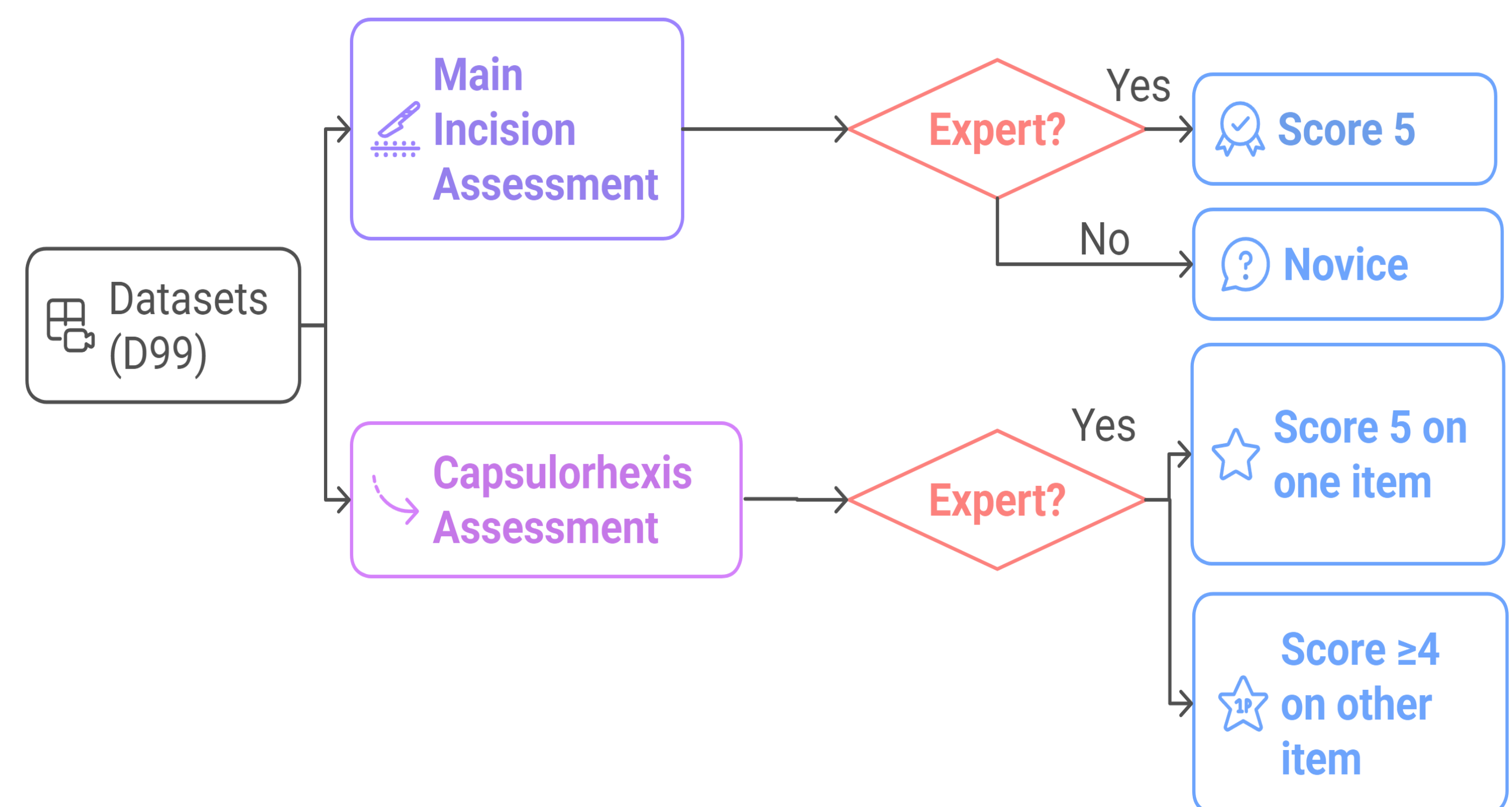
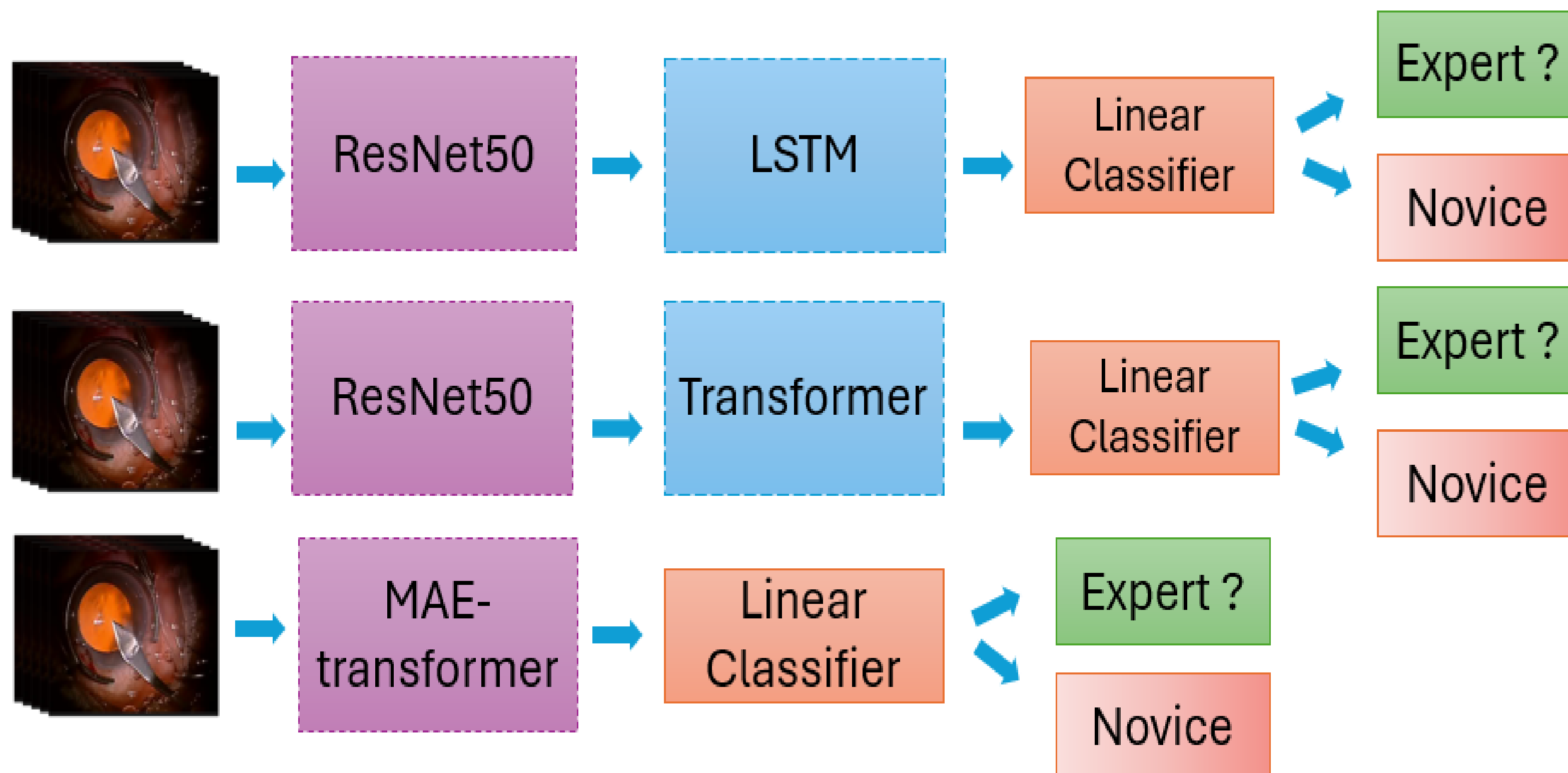
Cataract Surgery
is the Most
Performed
Surgery Globally



AI-Driven
Assessment and
Feedback on
Critical Steps can
improve
outcomes



METHODS



RESULTS

Main Incision

Models	Accuracy	AUC	Specificity	Sensitivity
CNN-LSTM	0.63 (0.44, 0.78)	0.64 (0.49 to 0.79)	0.36 (0.16, 0.61)	0.92 (0.67, 0.99)
CNN-ViT	0.62 (0.43, 0.79)	0.55 (0.45 to 0.65)	1 (0.78, 1.0)	0.1 (0.02, 0.4)
Video MAE-Transformer	0.76 (0.55, 0.89)	0.76 (0.57 to 0.95)	0.8 (0.49, 0.94)	0.73 (0.43, 0.9)

Capsulorhexis

Models	Accuracy	AUC	Specificity	Sensitivity
ResNet 101	0.62	0.45	0.80	0.76
STMAE	0.66	0.55	0.80	0.68
Video MAE-Transformer	0.71	0.55	0.90	0.85

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KEY TAKEAWAYS

Pre-training provides superior spatial-temporal representations for skill assessment, outperforming CNN-based architectures like CNN-LSTM and CNN-ViT.

Pre-training significantly enhances specificity and sensitivity for both main incision and capsulorhexis, surpassing previous methods.

AI models were tested using cross-validation, but further validation on new datasets is required to confirm generalizability.

Video MAE enables an objective, automated evaluation of critical surgical steps in cataract surgery, improving assessment accuracy.

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